



Predicting the progression of MCI and Alzheimer's disease on structural brain integrity and other features with machine learning

Marthe Mieling · Mushfa Yousuf · Nico Bunzeck · for the Alzheimer's Disease Neuroimaging Initiative

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Abstract Machine learning (ML) on structural MRI data shows high potential for classifying Alzheimer's disease (AD) progression, but the specific contribution of brain regions, demographics, and proteinopathy remains unclear. Using Alzheimer's Disease Neuroimaging Initiative (ADNI) data, we applied an extreme gradient-boosting algorithm and SHAP (SHapley Additive exPlanations) values to classify cognitively normal (CN) older adults, those with mild cognitive impairment (MCI) and AD dementia patients. Features included structural MRI, CSF status, demographics, and genetic data. Analyses comprised one cross-sectional multi-class classification (CN vs. MCI vs. AD dementia, $n = 568$) and two

longitudinal binary-class classifications (CN-to-MCI converters vs. CN stable, $n = 92$; MCI-to-AD converters vs. MCI stable, $n = 378$). All classifications achieved 70–77% accuracy and 61–83% precision. Key features were CSF status, hippocampal volume, entorhinal thickness, and amygdala volume, with a clear dissociation: hippocampal properties contributed to the conversion to MCI, while the entorhinal cortex characterized the conversion to AD dementia. The findings highlight explainable, trajectory-specific insights into AD progression.

Keywords Alzheimer's disease · Magnetic resonance imaging · Structural degeneration · Machine learning · Classification

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Marthe Mieling and Mushfa Yousuf contributed equally to this work.

M. Mieling (✉) · M. Yousuf · N. Bunzeck (✉)
Department of Psychology, University of Lübeck,
Ratzeburger Allee 160, 23562 Lübeck, Germany
e-mail: m.mieling@uni-luebeck.de

N. Bunzeck
e-mail: nico.bunzeck@uni-luebeck.de

N. Bunzeck
Center of Brain, Behavior and Metabolism (CBBM),
University of Lübeck, Ratzeburger Allee 160,
23562 Lübeck, Germany

Introduction

Structural brain integrity has long been associated with cognitive abilities across the adult lifespan [1–3]. While a distinction can be made between healthy vs. pathological changes in mild cognitive impairment (MCI) and Alzheimer's disease (AD) dementia, it is also clear that age-related changes can be described on a continuum rather than one-off events, and, therefore, understanding developmental trajectories is essential. In this regard, machine learning [4] offers novel ways to systematically investigate the contribution and developmental changes of complex feature constellations,

including brain regions, proteinopathy, or genetic factors; however, longitudinal analyses using machine learning with cognitively normal (CN) older adults, MCI, and AD dementia patients are scarce. Filling this apparent gap, and identifying fingerprints of complex feature constellations in the context of AD progression, is the goal of this work.

Apart from typical neurodegeneration, especially within medial temporal lobe regions, such as the hippocampus and entorhinal cortex, the progression of AD is associated with the accumulation of amyloid- β (A β) and hyperphosphorylated tau (pTau) [5–7]. These neuropathologies may all serve as biomarkers, but the initial stages of AD often manifest without noticeable symptoms, making it challenging to identify them early on [7]. Previous research indicates that the progression of AD is distinguished by the dissemination of structural brain degeneration, which is influenced by A β and pTau proteinopathy, quantified based on cerebrospinal fluid (CSF) assays. Accordingly, initial structural degeneration spreads from subcortical brain regions, such as the nucleus basalis of Meynert (NbM) but also possibly the locus coeruleus [8], to the entorhinal cortex moderated by CSF status [9, 10]. Importantly, this is followed by a spread of neural degeneration from the entorhinal cortex to temporal and parietal brain regions, which can be detected in vivo by structural MRI [9] even in preclinical AD. For instance, subjects with preclinical AD (i.e., cognitively normal and abnormal A β and pTau biomarkers) showed more atrophy in the entorhinal cortex than cognitively healthy (i.e., cognitively normal, normal A β , and pTau biomarkers) [9]. Hence, the development of AD can be described on a continuum encompassing normal aging, initial pathological shifts in biomarkers, followed by the onset of mild symptoms as seen in MCI, finally leading to the development of AD dementia [7]. In this regard, CSF status and MRI measures of structural brain integrity appear to be important biomarkers [5, 11].

Previous studies have already used different machine learning approaches to classify high-performing CN, MCI, and AD dementia patients based on structural MR images [12–14]. Classification accuracies in binary classifications ranged from 80 to 100% for CN vs AD dementia, 60–90% for CN vs. MCI, and 50–85% for AD dementia vs MCI [15, 16]. These effects were often based on

lower volumes in MCI and AD dementia as compared to CN in specific brain regions, which is in line with the prevailing view on the etiological and neuroscientific comprehension of AD dementia [17–21]. These include the hippocampus, amygdala, entorhinal cortex, precuneus, cingulate gyrus, and the rostral and caudal regions within the medial frontal lobe [22]. In particular, hippocampal and entorhinal integrity plays a crucial role in the progression and prediction of AD [13, 23, 24] since atrophy in these regions strongly predicts future cognitive impairment and serves as an early indicator of AD dementia [25–28]. Despite the apparent specificity, especially within the medial temporal lobe, AD encompasses widespread neurodegenerative processes throughout the entire brain [17], which have not fully been addressed. Along these lines, research on ternary classifications of CN vs. MCI vs. AD dementia, and longitudinal analyses using machine learning on the disease progression, involving the conversion from CN to MCI or MCI to AD dementia, are scarce or do not provide information on the contributing features (i.e., brain regions) of the classification [13–15, 29–31].

Therefore, we utilized data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) to examine how differences in brain integrity (volume and thickness) and additional features, such as CSF status, age, sex, education, and APOE4 genotyping, contribute to the classification of CN vs. MCI vs. AD dementia (cross-sectional, analysis 1, Fig. 1A). In a second step, we performed a longitudinal binary-class classification of CN older adults who later converted to MCI vs those that remained healthy (i.e., CN-to-MCI converters vs. CN stable, analysis 2, Fig. 1B), and MCI patients that later converted to AD dementia vs those that remained MCI (i.e., MCI-to-AD converters vs. MCI stable, analysis 3, Fig. 1C). All three analyses were based on state-of-the-art XGBoost (eXtreme Gradient Boosting) machine learning algorithm, and the assessment of SHAP (SHapley Additive exPlanations) values to pinpoint the contributing features for the classification models. We expected that the integrity of medial temporal lobe brain regions, especially the hippocampus and entorhinal cortex, contribute to the distinction of CN, MCI, and AD dementia, and, importantly, they would allow a precise prediction of the disease progression.

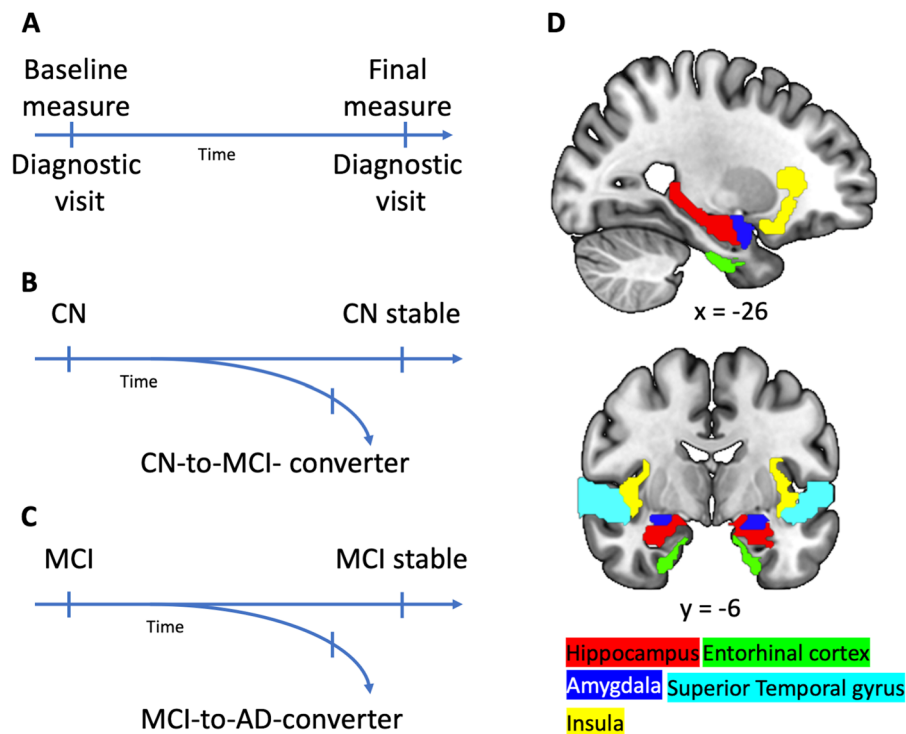


Fig. 1 Schematic illustration of the experimental design and analyses. All subjects underwent at least two diagnostic visits, including a baseline measure and follow-up (**A**). In a first cross-sectional analysis, cognitively normal (CN) older adults were compared against patients with mild-cognitive impairment (MCI) and patients with Alzheimer's disease dementia (AD). **B** depicts analysis 2 with CN that either remained stable over all follow-ups or developed a MCI at a following measure.

C Analysis 3 with MCI patients who remained stable and MCI patients who later converted to AD dementia. **D** Illustration of those brain regions that, across analyses, emerged as particularly relevant. Regions, except for the entorhinal cortex, were extracted based on the Automated Anatomical Labeling (AAL) atlas [32], while the entorhinal cortex was extracted using the Anatomy Toolbox [33–35] and overlaid onto a T1-weighted standard brain template

Methods

ADNI data

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). ADNI was launched in 2003 as a public–private partnership, led by Principal Investigator Michael W. Weiner, MD, to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to characterize the progression of mild cognitive impairment (MCI) and early Alzheimer's disease.

Participants and diagnoses

We included participants based on their diagnosis as CN, MCI, and AD dementia at the time point of MRI measurement. The assignment of diagnoses was conducted by the ADNI Clinical Core, which categorized individuals into three groups: CN (Clinical dementia rating [CDR] = 0, Mini Mental State Examination [MMSE] = 24–30), MCI (CDR = 0.5, MMSE = 24–30), and AD dementia (CDR = 0.5–1, MMSE = 20–26). These classifications are commonly used in clinical trials to assess cognitive and functional measures [36, 37] (for more information on the diagnostic procedure and in- and exclusion criteria, please see <http://adni.loni.usc.edu>).

For the first analysis (CN vs. MCI vs. AD dementia, Fig. 1A), the disease classification was based on a participant's baseline measures. In the longitudinal analyses 2 (CN-to-MCI converters vs CN stable, Fig. 1B) and 3 (MCI-to-AD converters vs MCI stable, Fig. 1C), for the converters (CN-to-MCI converters and MCI-to-AD converters) the latest available measurement as CN before the conversion to MCI (minimum 4 months), and AD dementia (minimum 4 months), respectively, was selected. Therefore, the term “longitudinal” refers to the diagnostic information rather than changes in MRI measures. For CN stable and MCI stable, the earliest available measurement was chosen (e.g., baseline information or if MCI converted from CN, then the first measurement associated with MCI), and time of stability was assessed by the time interval between the earliest available measurement and the last available ADNI visit. These participants remained stable for at least one measurement or longer and did not undergo conversion in their follow-up visits recorded by ADNI. For all analyses, participants with reverting diagnoses (e.g., MCI to CN, or AD dementia to MCI) were excluded.

Participants were only included with available high-quality FreeSurfer MRI, CSF drawings, and genotyping data acquired in close proximity with their neuropsychological assessment (< 6 months ago) (for more specific information regarding the criteria for inclusion and exclusion in ADNI, see <http://adni.loni.usc.edu>). Table S1 gives an overview of the participants' demographics, information on APOE4 genotype and harmonized CSF assay Table S2 shows neuropsychological test results.

MRI acquisition and FreeSurfer processing

The participants' MRI scans were obtained at various sites equipped with 1.5 and 3.0 Tesla scanners, following the standardized monitoring protocols of ADNI (see <http://adni.loni.usc.edu> for more information). Importantly, the available T1 images were already processed by ADNI (the Tosun laboratory at the University of California-San Francisco (see <https://adni.loni.usc.edu/data-samples/data-types/mri/>) using the cross-sectional pipeline [38] in FreeSurfer [39] (version 4.3, 5.1 and 6.0, <https://surfer.nmr.mgh.harvard.edu/>). FreeSurfer is an open-source software package for processing, analyzing, and visualizing brain images (<https://surfer.nmr.mgh.harvard.edu/>).

The features were extracted based on the Desikan-Killany atlas [40] and the subcortical whole brain segmentation [41, 42]. The tables provided by ADNI (versions 4.3, 5.1, and 6.0) were compiled by us, and the corresponding participants were selected accordingly.

In ADNI 1/GO, FreeSurfer version 4.3 encompassed data acquired from 1.5 Tesla scanners, version 5.1 included data from 3.0 Tesla scanners in ADNI GO/2, and version 6.0 integrated data from 3.0 Tesla scanners in ADNI 3. We merged FreeSurfer results, recognizing a potential bias [43]; however, FreeSurfer morphometric techniques have displayed strong test–retest reliability across various scanner manufacturers and field strengths [44]. Due to a limited number of participants in analyses 2 and 3, we combined data from 1.5 Tesla and 3.0 Tesla scanners, including all available FreeSurfer versions. In analysis 1, which had a sufficient number of participants, we adopted a more stringent approach by only including data from 3.0 Tesla scanners, utilizing the newer FreeSurfer versions 5.1 and 6.0. Further, we corrected for scanner manufacturer to mitigate bias and used only cross-sectional data.

The methods for identifying and calculating regional brain volume and thickness using FreeSurfer have been previously described [41]. Briefly, the volumes of brain regions were corrected for intracranial volume (ICV) by dividing the regions of interest (ROI) of the left and right hemispheres by the ICV. In the following, ICV-corrected bilateral ROI volumes were averaged; a similar approach was adopted for ROI thicknesses. Only MRI scans that exhibited exceptional overall segmentation and successfully passed ADNI's quality assessment were included. We only included FreeSurfer information with no missing values for each participant. In total, volume information on 46 bilateral brain regions and average thickness information on 34 bilateral brain regions (see Table S3) served as input for our XGBoost classification model.

CSF biomarker

AD's neuropathology involves the accumulation of A β , leading to plaques, and pTau, resulting in neurofibrillary tangles [45, 46]. We included CSF samples obtained through a fully automated Elecsys® protocol for A β and pTau measurements related to MRI

acquisition dates to comprehend their association with structural degeneration and diagnostic classification. A β 1–42 and pTau181 values were extracted for each participant. It is important to note that the Elecsys® protocols are still being developed, and the reported results are restricted to a specific technical limit (> 1700 pg/mL). Higher values beyond this limit were extrapolated from the calibration curve solely for research purposes, not diagnostics (for additional information on CSF draws and analyses see <http://adni.loni.usc.edu>).

In this study, we included both proteins in the classification models as a previously established pTau/A β ratio to depict the pathological processes of both proteins, demonstrating high concordance with PET measures and clinical diagnoses [47]. Analysis 1 involved CSF, analysis 2 was run without CSF due to the limited number of participants with CSF data ($n = 32$, Table S1), and analysis 3 was run with and without CSF since CSF data were available only for a limited number of participants ($n = 88$, Table S1).

APOE genotyping

The Apolipoprotein E (APOE) gene's $\epsilon 4$ allele is widely recognized as the most influential genetic risk factor for non-familial AD [48], and it is related to the grey matter atrophy in AD spectrum [49]. To better understand the diagnostic classification relying on structural ROIs, we incorporated APOE4 genotyping as a variable, factoring in the count of APOE4 alleles (0, 1 or 2) [50]. A blood sample was collected during the screening or baseline visit for APOE genotyping, and the analysis of genotypes followed standard procedures [51].

Neuropsychological assessment

The participants' cognitive performance was investigated using a comprehensive neuropsychological test battery. Here, we included the validated memory (MEM) and executive function assessments, as determined by confirmatory factor analysis [52, 53]. The memory scores encompassed the Alzheimer's Disease Assessment Scale, Logical Memory test, MMSE, and Rey Auditory Verbal Learning Test (RAVLT). The executive function scores were derived from the Category Fluency, Digit Span Backwards, Digit Symbol Substitution, Trails A and B, and Clock Drawing tests

[52, 53]. Furthermore, to gain a more comprehensive understanding of the participants' cognitive profiles, we also included the Montreal-Cognitive-Assessment (MoCA), the Sum of Boxes in the Clinical Dementia Rating Scale (CDRSB), and the Alzheimer's Disease Assessment Scale Cognitive (ADAS-Cog 13) (see specific details below).

Ensemble model: XGBoost

For binary and multi-class classification, XGBoost version 1.6.2 (<https://xgboost.readthedocs.io/en/latest/python/>) was used [54] in Python version 3.9.12 [55] and scikit-learn version 1.0.2 [56] on Jupyter Notebook version 6.4.12 [57] running on Windows 11. XGBoost is a gradient-boosting-based algorithm that uses decision trees as its base model [54], and it has been recently shown to perform well in various regression and classification tasks, including predicting brain age and classifying brain tumors [58–60]. Splitting data into training and testing sets is a common practice in machine learning to assess a trained model's performance on unseen test datasets. Here, the training and testing data sets were defined separately for each analysis. Specifically, XGBoost models were trained on 65% (training data set) of the stratified data and then evaluated on the remaining 35% (testing data set). A grid search was performed to optimize the hyper parameters of an XGBoost model by using a fivefold cross-validation, which included imaging data (volume and thickness), CSF, demographics (age and sex), and genotyping features as inputs. Optimizing hyperparameters aims to find the best set of hyper parameters resulting in high performing XGBoost models. Balanced accuracy was used as the scoring metric for multi-class classification analysis that comprises imbalanced data. Balanced accuracy accounts for imbalanced class distributions, i.e., it gives equal weight to each class regardless of the number of samples within each class [61, 62]. Furthermore, the scoring metric accuracy was used for binary classification analyses that comprise the balanced data. Finally, SHAP version 0.41.0 (<https://shap.readthedocs.io/en/latest/>) was used to calculate and visualize the contribution of each input feature to the XGBoost binary and multi-class model's prediction [63].

In machine learning, precision, recall, and F1-score are metrics used to evaluate the performance

of classification models. The F1 score, a metric combining precision and recall, is vital for evaluating classification models as it balances their ability to classify positive instances and minimize false positives correctly. Precision measures the accuracy of positive predictions, while recall evaluates the model's capability to capture all positive instances, collectively providing a comprehensive evaluation of model performance. Additionally, global accuracy, which assesses the overall correctness of classifications across all classes, offers an overall view of model effectiveness. The error rates were calculated as $1 - \text{global accuracy}$ [64–66]. Post hoc analyses were carried out on the whole sample of each analysis (including training and testing data).

ADNI dataset classification

We conducted three analyses based on the subjects' diagnostic status provided by ADNI; individual subjects were only included once in any of the analyses. In the first analysis, a cross-sectional multi-class classification was performed by concatenating subjects with normal cognition (CN), MCI, and AD dementia into one group ($n = 568$). The CSF values of each subject were included as input in the first multi-class classification analysis. In the second analysis, a longitudinal binary classification was performed by concatenating CN stable and CN-to-MCI converters into one group ($n = 92$); here, no CSF values were included as input due to the low number of available data (see above). Third, a longitudinal binary classification was carried out by concatenating the MCI stable and MCI-to-AD converters into one group; here, we performed one analysis with ($n = 88$) and one without CSF ($n = 378$). As mentioned above, all three analyses included brain volume and thickness, demographics, and genetics (APOE4).

The optimized hyperparameters for the first analysis were $\text{gamma} = 0.01$, $\text{learning_rate} = 0.05$, $\text{max_depth} = 3$, $\text{min_child_weight} = 3$, $\text{n_estimators} = 250$, $\text{colsample_bytree} = 1$, and $\text{subsample} = 0.7$; for the second analysis: $\text{gamma} = 0$, $\text{learning_rate} = 0.001$, $\text{max_depth} = 1$, $\text{min_child_weight} = 5$, $\text{n_estimators} = 150$, $\text{colsample_bytree} = 0.7$, and $\text{subsample} = 1$; the third analysis with CSF: $\text{gamma} = 0$, $\text{learning_rate} = 0.01$, $\text{max_depth} = 1$, $\text{min_child_weight} = 3$, $\text{n_estimators} = 200$, $\text{colsample_bytree} = 1$, and $\text{subsample} = 0.5$; and for the third without CSF:

$\text{gamma} = 0.3$, $\text{learning_rate} = 0.1$, $\text{max_depth} = 6$, $\text{min_child_weight} = 3$, $\text{n_estimators} = 50$, $\text{colsample_bytree} = 0.7$, and $\text{subsample} = 1$.

Ethics approval and consent to participate

Each center collecting data for ADNI provided an IRB (Institutional Review Board) approval and meets ADNI's requirements in accordance with the declaration of Helsinki. Informed consent was obtained from all ADNI participants (for more information at <http://adni.loni.usc.edu>). Further, the analyses presented here were approved by the local Ethics Committee of the University of Lübeck and carried out after ADNI's recommendations including the approval of the manuscript before submitting to a journal.

Results

Diagnoses and neuropsychological tests

For each of the three analyses, one-way ANCOVAs on the neuropsychological assessments, corrected for age, sex, and education (in years), were carried out. As expected, in cross-sectional analysis 1, significant differences emerged between CN, MCI, and AD dementia (all tests $p < 0.05/n$, $n = 7$ tests) with significantly lower cognitive performance in MCI and AD dementia (Table S2). In longitudinal analysis 2, the CN-to-MCI converters converted after, on average, 15 months ($SD = 8.35$) and the CN stables remained stable for, on average, 70.54 months ($SD = 47.94$). The CN-to-MCI converters scored lower on ADNI MEM ($F(1,87) = 9.88$, $p = 0.002$, Table S2) and ADAS-Cog 13 ($F(1,87) = 14.44$, $p < 0.001$, Table S2); however, for the other cognitive assessments, there were no significant differences ($p < 0.05/n$, $n = 7$ tests, Table S2). In longitudinal analysis 3, MCI-to-AD converters converted, on average, after 9.53 months ($SD = 5.85$), and the MCI stables remained stable for, on average, 52.5 months ($SD = 93.5$). The MCI-to-AD converters had lower scores in all neuropsychological assessments compared to MCI stable (Table S2).

ADNI dataset classification results

XGBoost model performances for each analysis are reported in Table 1. For analysis 1 (CN vs MCI vs AD

Table 1 XGBoost model performance for each analysis

	precision	recall	f1-score	Macro average	global accuracy of the model	Error Rate	N _{Test}
<i>Analysis 1 (cross-sectional, CN vs MCI vs AD, with CSF status)</i>							
CN	72%	81%	76%	71%	71%	29%	95
MCI	61%	50%	55%				68
AD	83%	83%	83%				36
<i>Analysis 2 (CN stable vs. CN-to-MCI converters, without CSF status)</i>							
CN stable	72%	81%	76%	76%	76%	24%	16
CN-to-MCI converters	80%	71%	75%				17
<i>Analysis 3 (MCI stable vs MCI-to-AD converters, with CSF status)</i>							
MCI stable	74%	88%	80%	77%	77%	23%	16
MCI-to-AD converters	83%	67%	74%				15
<i>Analysis 3 (MCI stable vs MCI-to-AD converters, without CSF status)</i>							
MCI stable	68%	74%	71%	70%	70%	30%	66
MCI-to-AD converters	72%	66%	69%				67

The precision, recall, f1-score, macro-average, global accuracy of the model, and N_{Test} (the number of subjects included in the testing set) or each analysis separately. Abbreviations: *CN*, cognitive normal; *MCI*, mild cognitive impairment; *AD*, Alzheimer's disease dementia; *CSF*, cerebro spinal fluid

dementia, Fig. 2A), the XGBoost model for the cross-sectional multi-class classification achieved an overall global accuracy (i.e., classification accuracy) of 71%, and the model's f1-scores were 76% for CN, 55% for MCI, and 83% for AD dementia. SHAP revealed the following features with the highest predictive value in the classification: CSF status, hippocampus volume, and entorhinal cortex thickness (Fig. 2B). While the hippocampus volume and entorhinal cortex thickness were significantly lower, the CSF status was significantly higher ($p < 0.05/n$, $n = 9$ tests) across the three groups in the entire dataset, as confirmed by post hoc t -tests or Mann–Whitney U tests (when normal distribution was not given; Fig. 2C, Table S4).

For analysis 2 (CN stable vs. CN-to-MCI converters, Fig. 3A), the XGBoost model for longitudinal binary-class classification achieved an overall global accuracy of 76%, and the model's f1-scores were 76% for CN stable and 75% for CN-to-MCI converters. SHAP revealed the following features with the highest predictive value in the classification task: hippocampus volume, insula thickness, and superior temporal gyrus thickness (Fig. 3B). These brain regions showed lower values (i.e., volume and thickness, respectively) in CN-to-MCI converters compared to CN stable (Fig. 3C) as confirmed by post hoc t -tests ($p < 0.05/n$, $n = 3$ tests) calculated on the entire sample (Table S4).

For analysis 3 (MCI stable vs MCI-to-AD converters, Fig. 4A), the XGBoost model for longitudinal binary-class classification achieved an overall global accuracy of 77%, and the model's f1-score were 80% for the MCI stable and 74% for the MCI-to-AD converters. SHAP revealed the following features with the highest predictive value in the classification task: entorhinal cortex thickness, CSF status, and amygdala volume (Fig. 4B). Again, the brain features showed lower and the CSF status higher values in MCI-to-AD converters compared to MCI stable, further supported by post hoc t -tests or Mann–Whitney U tests ($p < 0.05/n$, $n = 3$ tests) calculated on the entire sample (Fig. 4C, Table S4). Finally, the same XGBoost model for binary-class classification without CSF status (Fig. 5A) showed an overall global accuracy of 70%, and the model's f1-score was 71% for MCI stable and 69% for MCI-to-AD converters. SHAP revealed the following features with the highest predictive value in the classification task: entorhinal cortex thickness, amygdala volume, and cuneus volume (Fig. 5B). Again, these features showed lower values in MCI-to-AD converters compared to MCI stable, confirmed by post hoc t -tests or Mann–Whitney U tests on the entire sample ($p < 0.05/n$, $n = 3$ tests) (Fig. 5C, Table S4) (see Fig. 1D for a depiction of the most relevant brain regions across our analyses).

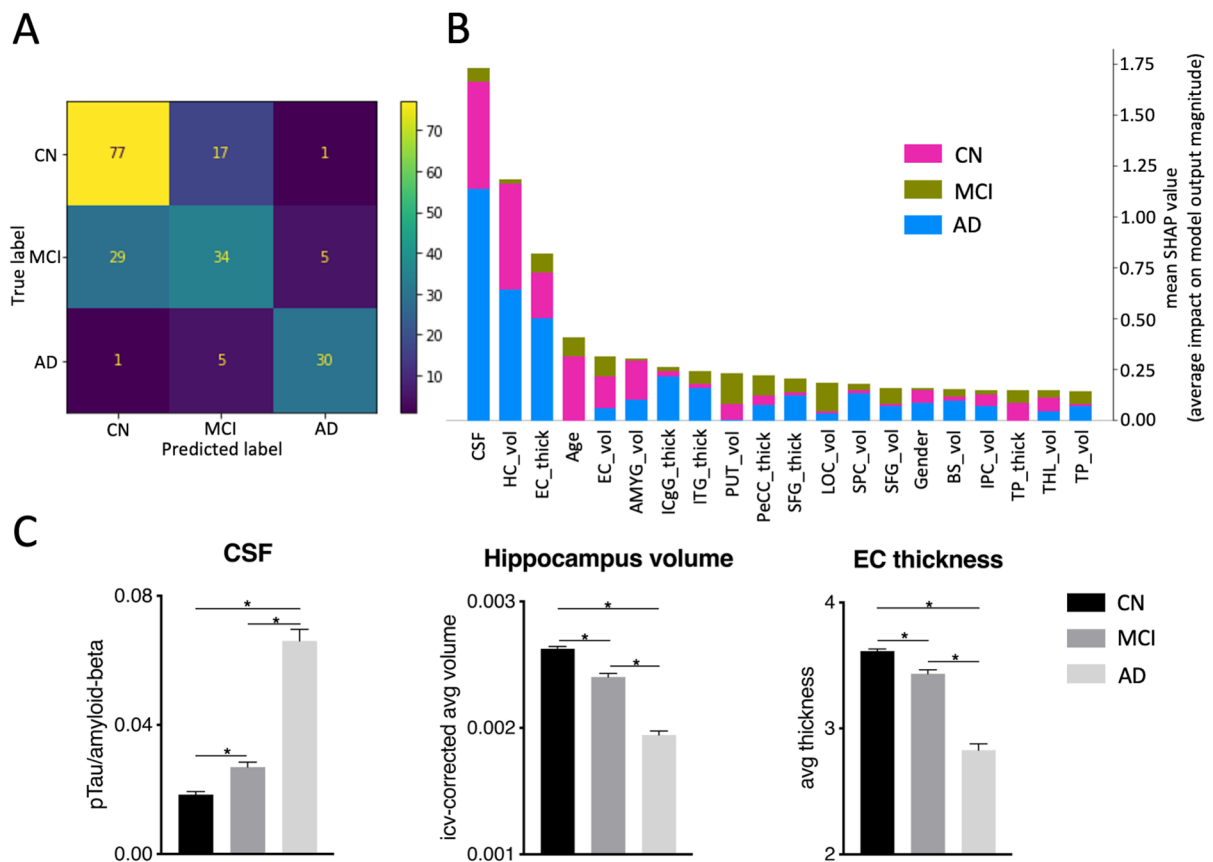


Fig. 2 Analysis 1, cross-sectional multiclass classification of CN vs MCI vs AD dementia. **A** Displays the confusion matrix of the testing data set ($n = 199$), visually representing the accuracy and misclassification of predictions among the groups. **B** SHAP plot highlighting key features and their respective impact magnitude on class predictions based on the testing data set. **C** Displays the three dominant features for the entire sample ($n = 568$): CSF status, hippocampus volume and entorhinal cortex thickness. Hippocampus volume and entorhinal thickness was significantly lower, and CSF ratio was significantly higher, across the three groups. *Significant after Bonferroni correction $p < 0.05/n$ ($n = 9$ tests).

Discussion

We investigated the underlying principles of AD progression on the basis of longitudinal diagnostic information, structural MRI data, APOE4, CSF status, and sociodemographics from healthy older adults, individuals with MCI and patients with AD dementia. Using a state-of-the-gradient-boosting based machine learning algorithm, we can show that a multi-class classification of CN vs. MCI vs. AD dementia is prominently driven by CSF status (i.e.,

Abbreviations: SHAP, SHapley Additive exPlanations; CSF, cerebrospinal fluid; HC_vol, hippocampus volume; EC_thick, entorhinal cortex thickness; AMYG_vol, amygdala volume; ICgG_thick, isthmus of cingulate gyrus thickness; ITG_thick, inferior temporal gyrus thickness; PUT_vol, putamen volume; PeCC_thick, pericalcarine cortex thickness; SFG_thick, superior frontal gyrus thickness; LOC_vol, lateral occipital cortex volume; SPC_vol, superior parietal cortex volume; SFG_vol, superior frontal gyrus volume; BS_vol, brain stem volume; IPC_vol, inferior parietal cortex volume; TP_thick, temporal pole thickness; THL_vol, thalamus volume; TP_vol, temporal pole volume

proteinopathy), hippocampal volume, and entorhinal thickness, which confirms previous work and demonstrates the feasibility of our approach. As a novel main observation, the accurate classification of a conversion from CN to MCI was mainly driven by hippocampal volume followed by thickness of the insula and superior temporal gyrus; the accurate classification of a conversion from MCI to AD dementia was primarily based on thickness of the entorhinal cortex followed by CSF status and amygdala volume. While this indicates a more prominent

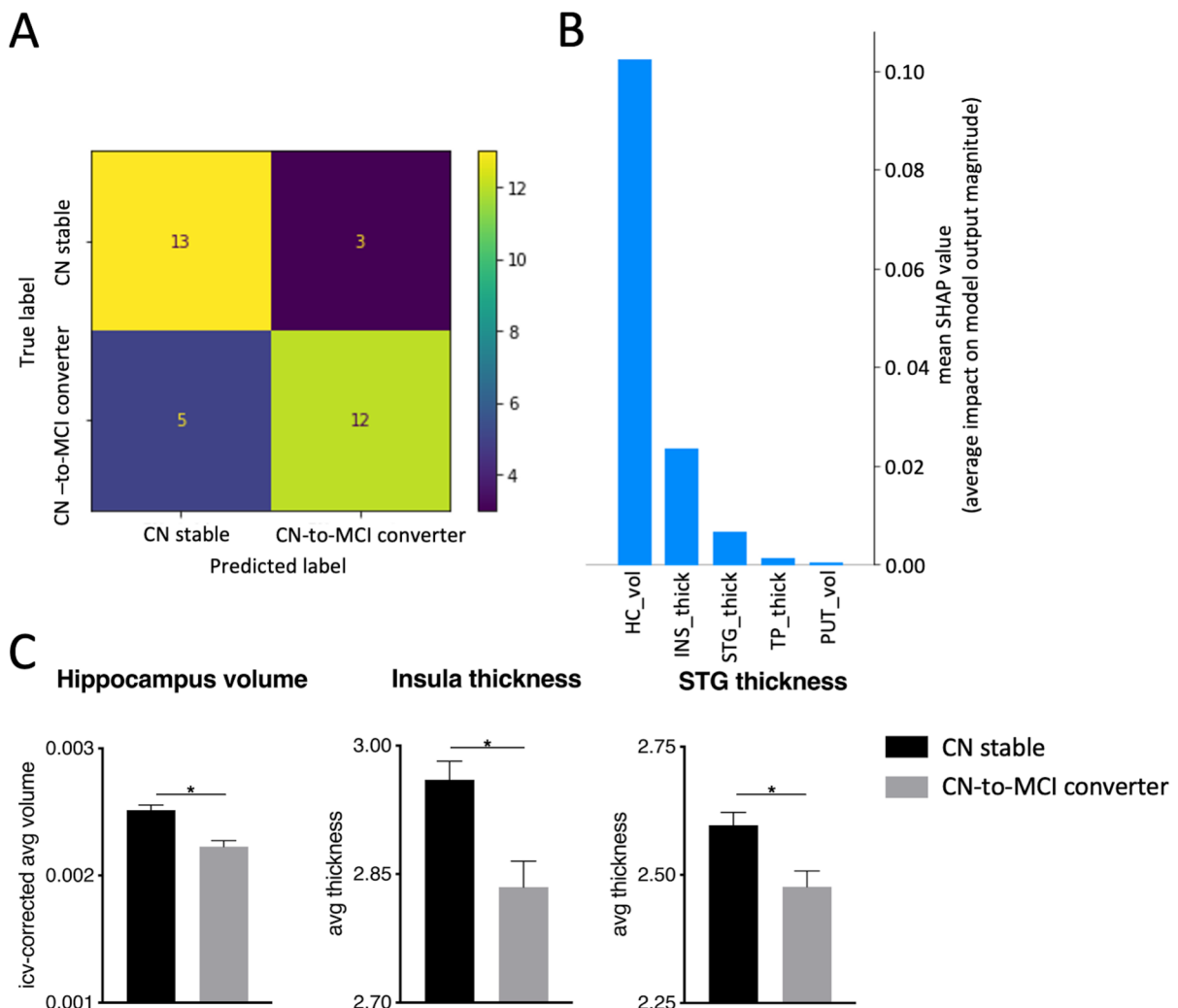


Fig. 3 Analysis 2, longitudinal binary-class classification on the testing data of CN stable vs CN-to-MCI converters. **A** Displays the confusion matrix, visually representing the accuracy and misclassification of predictions among the groups for the testing data set ($n = 33$). **B** SHAP plot highlighting key features and their respective impact magnitude on class predictions based on the testing data set. **C** Showing bar plots of the three dominant features for the entire sample ($n = 92$): hip-

pocampus volume; insula thickness; superior temporal gyrus (STG) thickness. All regions showed significantly lower values in the CN-to-MCI converter group. *Significant after Bonferroni correction $p < 0.05/n$ ($n = 3$ tests). Abbreviations: SHAP, SHapley Additive exPlanations; HC_vol, hippocampus volume; INS_thick, insula thickness; TP_vol, temporal pole volume; PUT_vol, putamen volume

role of the hippocampus in the conversion from CN to MCI and entorhinal cortex from MCI to AD dementia, our findings also demonstrate that, in comparison, other brain regions are less important for the classification. Therefore, our findings give novel insights into the developmental trajectories of AD, and they highlight novel ways to employ machine learning in the context of neuropsychiatric disorders.

Our cross-sectional multi-class classification of CN, MCI, and AD dementia achieved a global accuracy of 71%, which is comparable with previous research showing similar accuracies ranging from 59 to 77% [15–17]. Importantly, the three most relevant features of the classification were CSF ratio, hippocampal volume, and entorhinal thickness, followed by age and other features with comparably much lower predictive value (Fig. 2B). With regard

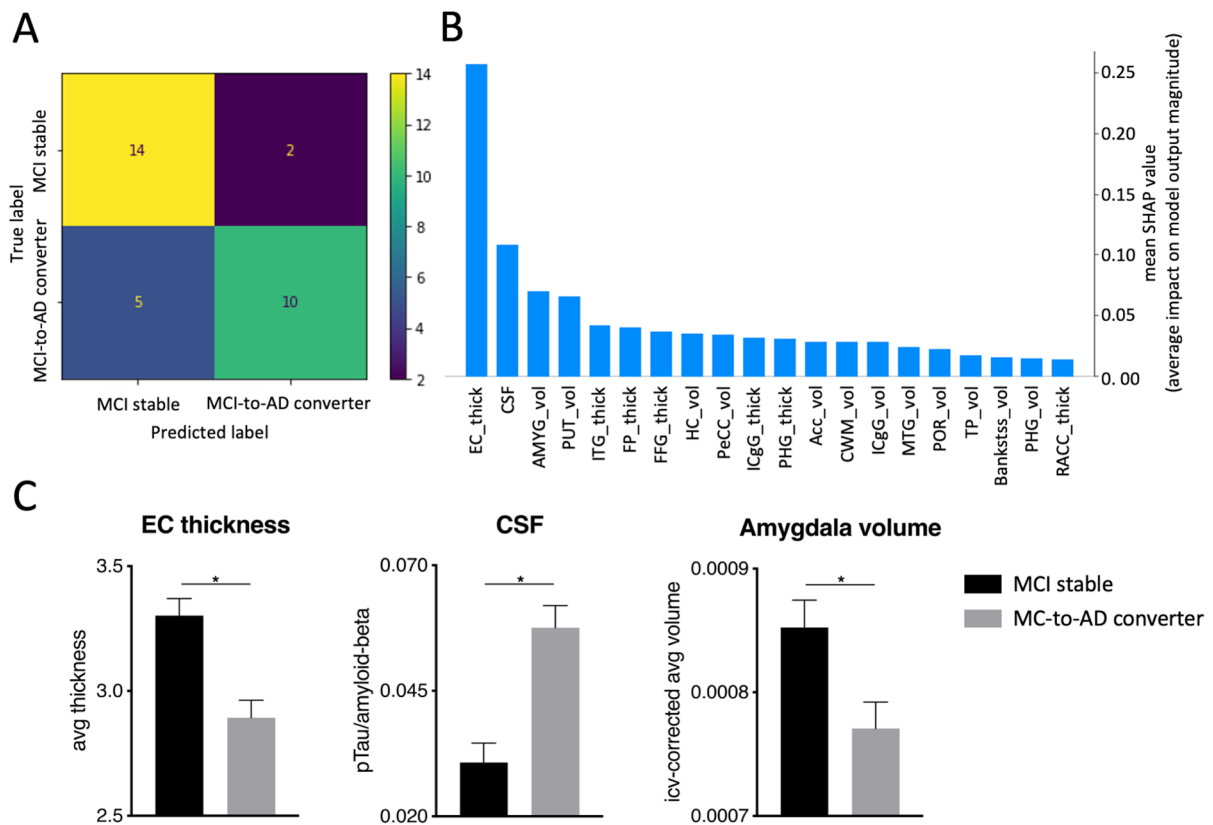


Fig. 4 Analysis 3, longitudinal binary-class classification on the testing data of MCI stable vs MCI-to-AD converters, including CSF status. **A** displays the confusion matrix of the testing data set ($n = 31$), visually representing the accuracy and misclassification of predictions among the groups. **B** features a SHAP plot highlighting key features and their respective impact magnitude on class predictions based on the testing data set. **C** shows the three dominant features in bar plots for the entire sample ($n = 88$): entorhinal cortex thickness, CSF status, and amygdala volume. Entorhinal cortex thickness and amygdala volume were significantly lower in MCI-to-AD converters, and the CSF ratio was significantly higher. *Significant after Bonferroni correction $p < 0.05/n$ ($n = 3$ tests). Abbreviations: SHAP, SHapley Additive exPlanations; CSF,

Cerebrospinal Fluid; EC_thick, entorhinal cortex thickness; AMYG_vol, amygdala volume; PUT_vol, putamen volume; ITG_thick, inferior temporal gyrus thickness; FP_thick, frontal pole thickness; FFG_thick, fusiform gyrus thickness; HC_vol, hippocampus volume; PeCC_vol, pericalcarine cortex volume; ICgG_thick, isthmus of cingulate gyrus thickness; PHG_thick, parahippocampal gyrus thickness; ACC_vol, accumbens area volume; CWM_vol, cerebellum white matter volume; ICgG_vol, isthmus of cingulate gyrus volume; MTG_vol, middle temporal gyrus volume; POR_vol, pars orbitalis volume; TP_vol, temporal pole volume; Bankstss_vol, banks of the superior temporal sulcus volume; PHG_vol, parahippocampal gyrus volume; RACC_thick, rostral anterior cingulate cortex thickness

to proteinopathy, A β and pTau have long been associated with AD [67], and individual CSF status can be superior compared to other features derived from structural MRI, FDG-PET, and APOE [50, 68]. This might be due to the fact that CSF biomarkers represent distinct biochemical aspects of AD pathology [69] and provide complementary information to structural MRI measures [50]. Therefore, abnormalities in CSF biomarkers, specifically A β , can precede neurodegeneration, including structural MRI changes

and cognitive impairment [5, 7], which is compatible with our observation of a reliable cross-sectional classification driven by CSF status [70]. Importantly, our classification approach (in all analyses) was based on a combination of several features, which appears to be the most reliable method of classifying the AD spectrum [50]. Along these lines, hippocampal volume and entorhinal cortex thickness were the second and third most prominent features that contributed to a reliable cross-sectional classification. Specifically,

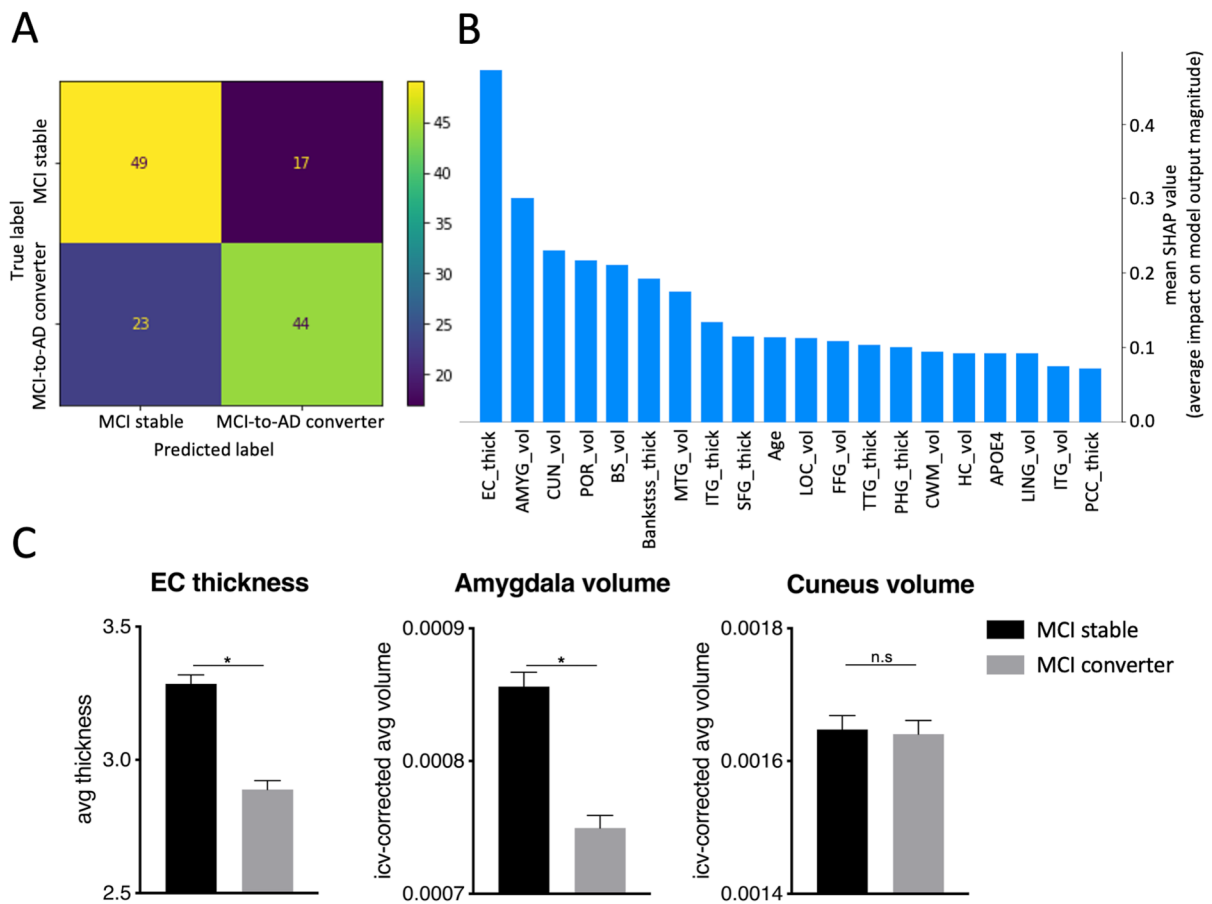


Fig. 5 Analysis 3, longitudinal classification of MCI stable vs MCI-to-AD converters, *not* including CSF. **A** displays the confusion matrix on the testing data ($n = 133$), visually representing the accuracy and misclassification of predictions among the groups. **B** SHAP plot highlighting key features and their respective impact magnitude on class predictions based on the testing data set. **C** shows the bar plots of the three dominant features: entorhinal cortex thickness, amygdala volume, and cuneus volume. Apart from the cuneus, all regions show significant reductions in MCI-to-AD converters. *Significant after Bonferroni correction $p < 0.05/n$ ($n = 3$ tests). Abbreviations: SHAP, SHapley Additive exPlanations; EC_thick, entorhinal

cortex thickness; AMYG_vol, amygdala volume; CUN_vol, cuneus volume; POR_vol, Pars orbitalis volume; BS_vol, brainstem volume; Bankstss_thick, banks of the superior temporal sulcus thickness; MTG_vol, middle temporal gyrus volume; ITG_thick, inferior temporal gyrus thickness; SFG_thick, superior frontal gyrus thickness; LOC_vol, lateral occipital cortex volume; FFG_vol, fusiform gyrus volume; TTG_thick, transverse temporal gyrus thickness; PHG_thick, parahippocampal gyrus; CWM_vol, cerebellum white matter volume; HC_vol, hippocampus volume; APOE4, apolipoprotein E4; LING_vol, lingual gyrus volume; ITG_vol, inferior temporal gyrus volume; PCC_thick, posterior cingulate cortex thickness

both brain regions showed more pronounced volume and thickness, respectively, in CN, followed by MCI and AD dementia, which resembles the disease progression (Fig. 2C, Table S4) [71, 72]. It also aligns with past studies using machine learning, in which hippocampal features and the entorhinal cortex strongly contribute to AD classification [17]. Finally, previous VBM studies comparing CN vs. AD dementia also revealed significant differences in grey matter

within the MTL, including hippocampus [73–75], entorhinal cortex [76–78], and amygdala [79, 80], which underlines the importance of medial temporal lobe brain regions in the development of AD, which will be discussed in more detail below.

Note that precision and recall in the cross-sectional multiclass classification of all three groups, MCI was associated with the lowest model's accuracy (f1-score of 55%, Table 1), followed by cognitively normal

older adults (76%, Table 1) and then AD dementia patients (83%, Table 1). This finding is compatible with previous reports of lower classifier performance for MCI [13, 15] and can be explained by the notion that MCI is clinically heterogeneous with distinct but less pronounced brain atrophy compared to AD dementia [13]. In a similar vein, it is important to note that our machine learning approach does not compare the effects of different features but combines them resulting in degrees of classifications.

Cross-sectional classifications can only portray group differences, but they cannot characterize longitudinal or developmental changes. Together with the fact that not all healthy older adults develop MCI, and not all MCI patients progress to AD dementia, it is important to understand the underlying and contributing factors [81]. Here, our longitudinal binary classification of CN-to-MCI converters vs. CN stable revealed the hippocampus volume in particular, but also the thickness of the insula and superior temporal gyrus (Fig. 3B), as the three most prominent features contributing to an accurate classification. In all three regions, healthy older adults that subsequently developed MCI showed a significantly lower volume and thickness, respectively, compared to those who remained CN and therefore did not develop MCI (Fig. 3C, Table S4). While this highlights the importance of the medial temporal lobe in the development of MCI, it is interesting to note that only the hippocampus but not the entorhinal cortex emerged as significant feature in our analysis [82, 83]. While this could be due to the rather low number of subjects in this analysis ($n = 46$ in each group), our classification model reflected a global accuracy of 76%, which is comparable with previous work reporting similar accuracies between 70 and 78% [43, 82, 83].

The contribution of the insula and superior temporal gyrus, with lower thicknesses in CN-to-MCI converters, is in line with cross-sectional VBM study results showing lower gray matter thickness in CN compared to MCI [84]. While cortical thinning originates in temporal brain regions [84–86] (see below) and is associated with neuropsychological performance [84], the role of the insula is less clear. However, as AD advances, the gyrification in the insular cortex decreases from MCI to more severe AD dementia, while folding complexity was linked to better memory performance specifically in AD dementia [87]. Furthermore, with AD progression the insula is

affected by proteinopathies [88] and structural degeneration, which impacts various functions such as olfaction [89], gustation, anatomic self-control, self-awareness, and emotion, possibly contributing to the diverse symptoms of AD [88, 90, 91].

Compatible with the observation of reduced hippocampal volumes, CN-to-MCI converters showed significantly worse memory functioning (MEM). Indeed, the hippocampus plays a central role in learning and memory with declines in structure and function even in healthy older adults [92]. While there were no significant differences in executive functioning or the MMSE, performances in other tests, however, point towards rather global differences in cognitive abilities (Table S2). For instance, the MoCA, which was lower in CN-to-MCI converters, quantifies several cognitive domains, including visuospatial/executive abilities, language and verbal abilities, memory as well as attention [93]. Importantly, both groups showed, on average, MoCA values higher than a cutoff value of 22 indicating their normal cognitive abilities [93]. Therefore, both groups are, from a clinical perspective, cognitively unimpaired, and they do not show significant brain atrophy; yet, healthy older adults who later develop MCI show lower grey matter volume, especially in the hippocampus, together with lower scores in several cognitive tests. While this observation does not allow a clear conclusion on whether structural degeneration precedes cognitive decline, which has been suggested before [5, 7], they clearly show that both factors—hippocampal volume and cognitive abilities—can provide significant biomarkers to predict developmental trajectories in cognitively normal older adults. Therefore, this finding is of high clinical relevance and adds novel insights to the limited number of studies using machine learning on the conversion from CN to MCI [43, 82, 83, 94, 95].

Our longitudinal binary classification with MCI patients who converted (MCI-to-AD converters) and MCI patients who remained stable (MCI stable) had global accuracies ranging from 70% (excluding CSF, Table 1) to 77% (with CSF, Table 1), which is comparable to others reporting accuracies ranging from 74 to 89% [96]. In our case, entorhinal cortex thickness and amygdala volume were the two most prominent features in both classification scenarios, with more pronounced reductions in MCI-to-AD converters compared to MCI stable (Fig. 4B and C and Fig. 5B

and C, Table S4). The entorhinal cortex is considered a critical relay for the communication between the neocortex and hippocampus, and, therefore, plays an important role in several cognitive processes, including learning and memory [97, 98]. With regard to AD progression, the entorhinal cortex appears to be one of the first brain regions affected, which could lead to memory impairments. For instance, deposits of A β and p-tau first occur in the trans-entorhinal and entorhinal cortex, which is followed by the hippocampus and other cortical regions [99]. MRI studies confirm this picture by demonstrating that volume [100] and shape [101] of the hippocampus as well as hippocampal subfields [102] differ as a function of disease progression (for a review see [103]). There are similar but more heterogeneous effects in the entorhinal cortex and the amygdala [104], and therefore, it is not surprising that both brain regions were the best predictors for the conversion from MCI to AD dementia in our study.

Interestingly, when analysis 3 was run solely on participants where CSF drawings were available, the entorhinal cortex still emerged as the most predictive feature followed by CSF status (Fig. 4B), which could have important implications for the clinical context. In fact, whole-brain MRIs are much less invasive than CSF drawings and bear a much lower risk of potential side effects. Physiologically, this possibly reflects a stronger atrophy of the entorhinal cortex as compared to CSF-based accumulations of proteinopathies in the disease progression. In other words, while A β and p-tau accumulations are particularly pronounced in the early stages, this could be followed by more pronounced neurodegeneration of the entorhinal cortex in the following stages from MCI to AD dementia, which is compatible with the AT(N) framework [7]. The AT(N) framework is based on A β (A), tau (T), and neurodegeneration (N) to categorize distinct disease stages of AD and to investigate the interaction of the pathological processes. However, cutoff values—often determined by PET and in some studies CSF—are not well defined [7] and large sample sizes are required to account for at least three groups (A-T-, A+ T-, A+ T+) [105]. While in our study the AT(N) framework could help to exclude other pathological conditions affecting the MTL (e.g., limbic-predominant age-related TDP-43 encephalopathy (LATE), primary age-related tauopathy (PART), Lewy body dementia (LBD), frontotemporal dementia (FTD)),

our work aligns more closely with clinical practice and ADNI by using neuropsychological assessments to diagnose MCI and AD dementia (see Methods).

The model without CSF status revealed the cuneus (Fig. 5B) as third most relevant contributing feature, which is in line with previous classifications of MCI-to-AD converters vs. MCI stable [106, 107]. Indeed, the cuneus is associated with processing visual information, integrating sensory information, and cognitive processes including attention, learning, and memory [108, 109], and its atrophy has been associated with an increased risk of AD [102]. However, in a post-hoc analysis, the cuneus did not show significant volume differences in MCI-to-AD converters compared to MCI-stable individuals. This might be explained by differences in the underlying statistical principles of both analyses and indicates the need for a careful interpretation.

Finally, we would like to point out the following strengths and limitations. First, our study included a rather large sample of well characterized subjects from the ADNI database. Some analyses, however, were based on smaller numbers (analysis 2 and analysis 3 with CSF assays available), which calls for cautious interpretations. Second, machine learning offers a robust and reliable approach facilitating generalization and optimization [110, 111]. In this regard, XGboost in combination with SHAP allowed us, in an agnostic way, to pinpoint the contributing features [63]. Note, however, that other approaches exist but a direct comparison was beyond the scope of this study (see [112] for more information). Third, our volumetric and thickness data were based on the well-established 2010 Desikan-Killiany atlas [40] and a subcortical segmentation [41, 42]. However, this does not include a precise quantification of some brain regions [41, 42], such as the nucleus basalis of Meynert (NbM), which might play a central role in the progression of AD [9, 10, 81, 113]. Fourth, ADNI data was measured at multiple sites with diverse MRI scanners, which we addressed by only pre-selecting high-quality data and including scanner manufacturers as covariates in our analyses. Fifth, CSF assays offer an important way to investigate the underlying neurobiology of AD, which, in future work, could be complemented by more regionally specific PET data. Sixth, we focused on participants with stable diagnoses and information on the conversion (with rather short conversion times), omitting potential reversions

[114]. Seventh, a model with more features, such as clinical, cognitive, genetic, PET, and CSF status, could enhance model accuracies [115], but this would require more data points (i.e., subjects). These aspects, together with alternative but not mutually exclusive frameworks [7], should be considered in future work.

Conclusion

To conclude, machine learning in combination with structural MRI data, sociodemographic, CSF status, and genetic information allows a precise and explainable classification along the AD continuum. Our findings indicate that the hippocampus plays a prominent role in the conversion from healthy aging to MCI; the entorhinal cortex, on the other hand, contributes more to the conversion from MCI to AD dementia. As such, our study gives novel insights into the developmental trajectories from healthy to pathological aging by suggesting a dissociation of specific medial temporal lobe brain regions. This, in turn, could contribute to earlier and more accurate diagnoses and interventions.

Abbreviations *Aβ*: Amyloid-β; *APOE*: Apolipoprotein E; *ADAS-Cog*: Alzheimer's Disease Assessment Scale Cognitive; *ADNI*: Alzheimer's Disease Neuroimaging Initiative; *CDRSB*: Clinical Dementia Rating Scale; *CN*: Cognitive normal; *ICV*: Intracranial volume; *MCI*: Mild cognitive impairment; *MEM*: Memory; *MMSE*: Mini-Mental-State Examination; *MoCA*: Montreal-Cognitive-Assessment; *NbM*: Nucleus basalis of Meynert; *pTau*: Hyperphosphorylated Tau; *RAVLT*: Rey Auditory Verbal Learning Test; *ROI*: Region of interest; *SHAP*: SHapley Additive exPlanations; *XGBoost*: EXtreme Gradient Boosting

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I. ADNI 1, GO, 2, 3, 4
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 Li Shen, PhD University of Pennsylvania.
 ADNI4 Amyloid PET Visual Read Team
 David Soleimani-Meigooni, MD University of California, San Francisco.
 Renaud La Joie, PhD University of California, San Francisco.
 Konstantinos Chiotis, MD, PhD University of California, San Francisco.
 Maison Abu Raya, MD University of California, San Francisco.
 Agathe Vrillon, MD, PhD University of California, San Francisco.
 Charles Windon, MD University of California, San Francisco.
 Julien Lagarde, MD, PhD University of California, San Francisco.
 Zoe Lin University of California, San Francisco.
 Aidyn Rose Hills University of California, San Francisco.
 ADNI4 Amyloid Disclosure Team
 Jason Karlawish, MD University of Pennsylvania.
 Claire Erickson, PhD University of Pennsylvania.
 Joshua Grill PhD University of California, Irvine.
 Emily Largent PhD University of Pennsylvania.
 Kristin Harkins MPH University of Pennsylvania.
 Early Project Development
 Michael W. Weiner, MDUCSF/NCIRE.
 Leon Thal, MD – Past Investigator.
 Zaven Khachaturian, PhD Khachaturian, Radebaugh & Associates (KRA), Inc.
 Richard Frank, MD, PhD General Electric.
 Peter J. Snyder, PhD University of Connecticut Alzheimer's Association's Ronald and Nancy Reagan's Research Institute.
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- Luter Liu.
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 Romina Romero, PhD Steven Tam, MD.
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 BMSc, ARRT(R)(MR) Sarah Malakauskas, MS.
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 Dahlgren, BS Alexandria Montero, RN.
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 Ruby Endre, M.R.T Zhala Taghi-Zada.
 University of British Columbia Clinic for AD & Related
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 Robin Hsiung, MD Carolyn English Ellen Kim, BA Eugene
 Yau.
 Haley Tong.
 Laura Barlow, RTR/RTMR Lauren Jennings.
 Michele Assaly Paula Nunes, PhD Tahlee Marian.
 Cognitive Neurology St. Joseph's Ontario:
 Andrew Kertesz, MD John Rogers, MD Dick Trost, PhD.
 Cleveland Clinic Lou Ruvo Center for Brain Health
 Dylan Wint, MD Charles Bernick, MD Donna Munic, PhD.
 Northwestern University:
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 Caila Ryan, MS Jelena Pejic.
 Kailey Basham, BS Leena Lukose, BS Loreece Haddad,
 MS.
 Lucas Quinlan, BS, MLS (ASCP) Nathaniel Houghtaling.
 Premiere Research Inst (Palm Beach Neurology):

Carl Sadowsky MD Walter Martinez MD Teresa Villena MD.

Georgetown University Medical Center:

Brigid Reynolds, NP Angelica Forero, MS Carolyn Ward, MSPH Emma Brennan, BS Esteban Figueroa Giuseppe Esposito, MD Jessica Mallory.

Kathleen Johnson, RN, NP Kathryn Turner, BSN Katie Seidenberg.

Kelly McCann, BA Margaret Bassett, NP Melanie Chadwick, NP.

Raymond Scott Turner, MD, PhD Robin Bean, RT.

Saurabh Sharma, MD.

Brigham and Women's Hospital:

Gad Marshall, MD Aferdita Haviari, BA Alison Pietras, PA-C, ACP Bradley Wallace, BS Catherine Munro, PhD.

Gladiliz Rivera-Delpin, MA Hadley Hustead, BS Isabella Levesque.

Jennifer Ramirez, BA Karen Nolan, BS, RT (MR).

Kirsten Glennon, RN, CNRN Mariana Palou, BA.

Michael Erkinen, MD.

Nicole DaSilva.

Pamela Friedman, Psy. D Regina M. Silver, RN Ricardo Salazar, MD Roxanne Polleys, AA Scott McGinnis (094), MD Seth Gale, MD.

Tia Hall, BS Tuan Luu.

Stanford University:

Steven Chao, MD Emmeline Lin, BS Jaila Coleman, BA.

Kevin Epperson, RT(R)(MR) Minal Vasanaawala.

Banner Sun Health Research Institute

Alireza Atri, MD, PhD Amy Rangel.

Brittani Evans Candy Monarrez Carol Cline, LMSW.

Carolyn Liebsack, RN, BSN, CCRC Daniel Bandy.

Danielle Goldfarb, MD Debbie Intorcchia Jennifer Olgin.

Kelly Clark.

Kelsey King, CCRP Kylee York.

Marina Reade, RN, FNP-C Michael Callan.

Michael Glass.

Michaela Johnson, G-ACNP, BC Michele Gutierrez.

Molly Goddard.

Nadira Trncic, MD, PhD Parichita Choudhury, MD Priscilla Reyes.

Serena Lowery Shaundra Hall Sonia Olgin.

Stephanie de Santiago, RN, NP.

Boston University:

Michael Alosco, PhD Alyssa Ton, BS.

Amanda Jimenez, MS, EMT-B, CPT Andrew Ellison, MR Technologist Anh Tran, RN.

Brandon Anderson, RT(N), CNMT Della Carter, MS.

Donna Veronelli, RTN, CNMT Steven Lenio, MD.

Eric Steinberg, RN, MSN, CNP Jesse Mez, MD, MS.

Jason Weller, MD Jennifer Johns, RN Jesse Mez, MD, MS Jessica Harkins, CNMT Alexa Puleio, MS.

Ina Hoti, BS.

Jane Mwicigi, MBChB., MPH Alexa Puleio, MS.

Michael Alosco, PhD Olivia Schultz, BA Mona Lauture, RN Eric Steinberg Ridiane Denis, RN Ronald Killiany, PhD Sarab Singh, CNMT Steven Lenio, MD Wendy Qiu, MD, PhD Ycar Devis, MPH.

Howard University:

Thomas Obisesan, MD, MPH Andrew Stone, MS.

Debra Ordor, RN, BSN Ifreke Udodong, CRNP.

Immaculata Okonkwo, DNP, MSN, APRN, FNP-BC Javed Khan, MD.

Jillian Turner, BS, MS Kyliah Hughes, BS, RMA Oshoze Kadiri, MPH.

Case Western Reserve University:

Charles Duffy, MD, PhD.

Ariana Moss.

Katherine Stapleton, LPN Maria Toth (fmr Gross), RN Marianne Sanders, BSN, RN Martin Ayres.

Melissa Hamski Parianne Fatica, CCRC Paula Ogrocki, PhD Sarah Ash.

Stacy Pot.

University of California, Davis Sacramento:

Doris Chen, MD Andres Soto.

Costin Tanase, PhD David Bissig, MD, PhD Hafsanoor Vanya, BA.

Heather Russell (116), CNMT Hitesh Patel, CNMT Hongzheng Zhang, CCRP Kelly Wallace, CCRP.

Kristi Ayers, BS Maria Gallegos, BS Martha Forloines, PhD Meghan Sinn.

Queenie Majorie S Kahulugan, CCRC Richard Isip, RT (R)(N)(CT).

Sandra Calderon, MS, RN, FMP-C Talia Hamm, BA, CCRP.

Parkwood Hospital: Michael Borrie, MD T-Y Lee, PhD

Dr Rob Bartha, PhD.

University of Wisconsin: Sterling Johnson, PhD Sanjay Asthana, MD Cynthia M. Carlsson, MD

Banner Alzheimer's Institute:

Allison Perrin, MD Pierre Tariot, MD Adam Fleisher, MD Stephanie Reeder, BA.

Dent Neurologic Institute Horacio Capote, MD Allison Emborsky

Anna Mattle, PharmD, MS Bela Ajtai, MD.

Benjamin Wagner, PA-C Bennett Myers.

Daryn Slazyk.

Delaney Fragale, PA-C Erin Fransen, PA Heather Macnamara Jonathan Falletta, PA-C Joseph Hirtreiter, RN Laszlo Mechtler, MD Megan King.

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Scott Wisniewski Stephanie O'Malley, PA-C Tatiana Jimenez-Knight Todd Peehler.

Traci Aladeen, PharmD Vernice Bates.

Violet Wenner Wisam Elmalik, MD.

Ohio State University:

Douglas W. Scharre, MD Arun Ramamurthy, MD Soumya Bouchachi, MD.

Maria Kataki, MD, PhD—Past Investigator Rawan Tarawneh, MD—Past Investigator Brendan Kelley, MD—Past Investigator.

Albany Medical College: Dzintra Celmins, MD Alicia Leader

Chris Figueroa Heather Bauerle, NP Katlynn Patterson Michael Reposa Steven Presto.

Tuba Ahmed.

Wendy Stewart.

Hartford Hosp, Olin Neuropsychiatry Research Center:

Godfrey D. Pearlson MD Karen Blank, MD.

Karen Anderson, RN.

Dartmouth-Hitchcock Medical Center:

Robert B. Santulli, MD Eben S. Schwartz, PhD.
Wake Forest University Health Sciences:
Jeff Williamson, MD, MHS, FACP Alicia Jessup, RN.
Andrea Williams Crystal Duncan.
Abigail O'Connell, APRN, FNP-C Karen Gagnon.
Ezequiel Zamora James Bateman.
Freda Crawford, CNMT Deb Thompson.
Eboni Walker Jennifer Rowell Mikell White, MHA.
Phillip "Hunter" Ledford Sarah Bohlman, MSL Susan Henkle, RN Joseph Bottoms, CNMT.
Lena Moretz, RT(R) CT (MR) Bevan Hoover, BS.
Michael Shannon Samantha Rogers, PA-C Wendy Baker.
William Harrison, MD.
Rhode Island Hospital: Chuang-Kuo Wu, MD Alexis DeMarco, BS
Ava Stipanovich, BS, ScM.
Daniel Arcuri, CNMT, RT(N)(CT) Jan Clark, RN, BSN, CCRC, CSNT.
Jennifer Davis, PhD Kerstin Doyon, RN, BSN.
Marie Amoyaw, BA.
Mauro Veras Acosta, PENDING, BS Ronald Bailey, RT-R, CNMT.
Scott Warren, MD Terry Fogerty.
Victoria Sanborn, PhD.
Butler Hospital
Meghan Riddle, MD Stephen Salloway, MD, MS Paul Malloy, PhD.
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Juris Janavs.
University of Chicago Emily Roglaski, PhD Meagan Yong Rebecca Devine
Eastern Virginia Medical School
Hamid Okhravi, MD.
Charter Health Research Services
Edgardo Rivera, MD Teresa Kalowsky Caroline Smith Christina Rosario.
Houston Methodist Neurological Institute
Joseph Masdeu, MD, PhD Richard Le, PharmD Maushami Gurung.
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Micah Ellis Slaughter Nadeen Elayan Skieff Acotley.
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Chelsea Reichert.
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Vanderbilt University Medical Center
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University of Texas Health, San Antonio
Sudha Seshadri, MD Amy Saklad.

Floyd Jones.
Rutgers University
William Hu, MD, PhD.
V. Sotelo.
Gonzalez & Aswad Health Services
Yaneicy Gonazalez Rojas, MD.
Medical University South Carolina
Jacob Mintzer, MD, MBA Crystal Flynn Longmire, PhD
Kenneth Spicer, MD, PhD.

Author contribution Marthe Mieling: conceptualization, data curation, project administration, methodology, formal analysis, writing—original draft. Mushfa Yousuf: conceptualization, data curation, methodology, visualization, formal analysis, writing—original draft. Nico Bunzeck: supervision, conceptualization, methodology, project administration, writing—original draft.

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Data availability All data from this study can be obtained freely upon request from the Image and Data Archive (IDA) operated by the Laboratory of Neuro Imaging (LONI) at <https://ida.loni.usc.edu>. Our analysis code is available at OSF: https://osf.io/ajwct/?view_only=4f2012b70a3b4ea18ac85e847767657d.

Declarations

Conflict of interest The authors declare no competing interests.

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